

Commuting things...

Monday, February 2, 2026 9:38 AM

Applied mathematics:

- ① Solve linear equations
- ② Use the triangle inequality
- ③ Commute things first, ask questions later

① $\sup_b \inf_a g(a,b) \leq \inf_a \sup_b g(a,b)$
any kind of r.v., any support, any g

proof: $\forall a, \forall b \quad g(a,b) \leq \sup_{b'} g(a,b')$

so $\forall b \quad \inf_a g(a,b) \leq \inf_a \sup_{b'} g(a,b')$

so $\sup_b \inf_a g(a,b) \leq \inf_a \sup_{b'} g(a,b')$

Ex showing it's strict: $g(a,b) = -(a-b)^2$

support: $a, b \in \{\pm 1\}$, a.u.nif. (± 1)

$\inf_a g(a,b) = -2 \forall b$ but $\sup_b g(a,b) = 0 \forall a$

so $-2 < 0$

② $\sup_b \inf_a g(a,b) \leq \inf_a \sup_b g(a,b)$

proof: same as ①

if "=" it's a "saddle pt."

Used in optimization: weak duality

Lots of study on conditions when it is an equality

③ $\lim_a \lim_b g(a,b) \stackrel{?}{=} \lim_b \lim_a g(a,b)$

No, Not "=" unless you have further conditions.

No " $\leq$ " version (though see Fatou's lemma)

Ex: $g(a,b) = 2^{a-b}$, $a, b \rightarrow \infty$ $0 \neq \infty$

OR $g(a,b) = (1 - \frac{1}{a})^b$, $a, b \rightarrow \infty$, $0 \neq 1$

④ $\min_a \min_b g(a,b) \stackrel{?}{=} \min_b \min_a g(a,b)$

Yes, though may need to be careful w/ $\inf$ vs. $\min$ in some cases

so write $\min_{a,b} g(a,b)$

Same holds true w/ $\max_a \max_b \dots$

⑤ $\int_a \int_b g(a,b) db da \stackrel{?}{=} \int_b \int_a g(a,b) da db$

Often, eg. under conditions for Fubini's Thm

⑥ $\lim_a \int_b g(a,b) db \stackrel{?}{=} \int_b \lim_a g(a,b) db$

and

$\frac{d}{da} \int_b g(a,b) db \stackrel{?}{=} \int_b \frac{\partial}{\partial a} g(a,b) db$

Sometimes... see more details on next pages

Fubini's Thm

If $\int_{X \times Y} |f(x,y)| dx dy < \infty$ then $\int_X \left(\int_Y f(x,y) dy \right) dx = \int_Y \left(\int_X f(x,y) dx \right) dy$
guaranteed if, eg., f cts and X, Y compact

Fatou's Lemma

If $f_n \geq 0$ are measurable then if $f(x) := \liminf_{n \rightarrow \infty} f_n(x)$, f is measurable and

$\int f \leq \liminf_{n \rightarrow \infty} \int f_n$

Ex: $f_n = \begin{cases} \frac{1}{n} & 0 \leq x \leq n \\ 0 & x > n \end{cases}$, $f(x) = 0$, $\int f_n = 1$
 $0 \leq 1 \checkmark$

SGD - commuting gradient and expectation

Wednesday, April 17, 2024 11:25 AM

Let $L_D(\omega) = \mathbb{E}_{z \sim D} l(\omega, z)$. Is $\frac{d}{d\omega} L_D(\omega) = \mathbb{E}_{z \sim D} \frac{\partial}{\partial \omega} l(\omega, z)$?

Answer: Sometimes

Example $z \sim \text{Unif}(\{\pm 1\})$, $l(\omega, 1)$ isn't differentiable, $l(\omega, -1) := -l(\omega, 1)$
so $L_D(\omega) = 0$ which is differentiable

Example let $L_D(\omega) = |\omega|$ for $\omega \in (-\pi, \pi)$

the Fourier series is $= \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos((2n-1)\omega)$

so

$$\frac{\pi}{2} - |\omega| = \frac{4}{\pi} \cdot c \cdot \mathbb{E}_{z \sim D} \cos((2z-1)\omega)$$

not differentiable
at $\omega=0$

where $P[z=n] = \frac{1}{c \cdot (2n-1)^2}$ c a normalization constant

$l(\omega, z) = \cos((2z-1)\omega)$ is differentiable.

OR...

use $L_D(\omega) = \omega$ which has Fourier series $\frac{1}{\pi} \sum_{n=-\infty}^{\infty} \frac{\sin^2(n\omega)}{n^2}$ $\omega \in (0, \pi)$
 $\uparrow$ not periodic $\uparrow$ periodic
... so discontinuities at $0, \pi, \dots$

OR... $L_D(\omega) = \frac{\omega^2}{4} - \frac{\pi}{2}\omega + \frac{\pi^2}{6}$, Fourier series $\sum_{n=1}^{\infty} \frac{\cos(n\omega)}{n^2}$ $\omega \in [0, 2\pi]$

Thm (1D case) cf. Folland Thm 2.2.7

Let $l: \mathcal{H} \times \mathcal{Z} \rightarrow \mathbb{R}$ where $\mathcal{H} = [a, b]$. If $\forall \omega \in \mathcal{H}, \mathbb{E} |l(\omega, z)| < \infty$

and $\frac{\partial l}{\partial \omega}$ exists, and $\exists g: \mathcal{Z} \rightarrow \mathbb{R}$ st.

$(\forall \omega, z) \quad \left| \frac{\partial l}{\partial \omega}(\omega, z) \right| \leq g(z)$ then

1) $L_D(\omega) := \mathbb{E}_{z \sim D} l(\omega, z)$ is differentiable, and

2) $\frac{d}{d\omega} L_D(\omega) = \mathbb{E}_z \frac{\partial l}{\partial \omega}(\omega, z)$

Proof: Lebesgue's Dominated Convergence Thm

More details on differentiating under the integral sign

Monday, April 28, 2025 3:09 PM

Kim Border's notes are cached here:

<https://healy.econ.ohio-state.edu/kcb/Ma103/Notes/DifferentiatingAnIntegral.pdf>

Integrals

$$f(x) = \int_{\Omega} \varphi(x, z) dz$$

c.f. Kim Border's notes for Ma 3 '20 at Caltech
"Supplement 4: Differentiating under an integral sign"

Theorem (informal) (refs: Border, or Aliprantis + Burkinshaw)

Assume $\forall x, \varphi(x, \cdot) \in L^1(\Omega)$, $\frac{\partial}{\partial x} \varphi(x, \cdot)$ exists and is also in L^1
i.e., $\int_{\Omega} |\varphi(x, z)| dz < \infty$

refs: Aliprantis + Burkinshaw
p. 193-194

and assume a **uniform local integrability condition**

(sufficient: Ω is bounded, $\frac{\partial}{\partial x} \varphi$ is jointly continuous)

or
Dudley, Appendix A
p. 417-423
"uniform CLTs" 2nd ed.

then f is differentiable and

$$\frac{d}{dx} f(x) = \int \frac{\partial}{\partial x} \varphi(x, z) dz$$

refs: Dieudonné p. 177
Thm 8.11.2, 1969
"Foundations of Modern Analysis"

Counterexamples (when assumptions not met)

in Gelbaum + Olmsted, p. 123 Ex. 9.15

"Counterexamples in Analysis" 2003, 1965

variant

$$f(x) = \int_{a(x)}^{b(x)} \varphi(x, z) dz$$

Theorem Leibniz Integral Rule (refs: wikipedia)

$x \in \mathbb{R}^1, z \in \mathbb{R}^1$

Assume φ and $\frac{\partial}{\partial x} \varphi$ are jointly continuous in (x, z) and
 $a(x), b(x)$ continuously differentiable, then

$$\frac{d}{dx} f(x) = \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} \varphi(x, z) dz + \varphi(x, b(x)) \cdot b'(x) - \varphi(x, a(x)) \cdot a'(x)$$

i.e., Fundamental Theorem of Calculus

Generally, use Lebesgue's Dominated Convergence Theorem to prove these results